

I'm parameterizing the world's most sophisticated cruise ship outbreak model and we cover 10 pathogens. I normalize norovirus, for example, against the VSP - and I realize that nobody has ever done it recognizing that the public health response is part of the observation. Meanwhile, there are a spectrum of infections and the VSP reported 'attack rate' is clinical infections only. So asymptomatic and subclinical are not reported (though people are encouraged to introspect during the interviews, i think). Can you give me the best information available (serological, molecular tests, etc) that might help me better parameterize the model?

Use Asymptomatic Fractions to Calibrate VSP

For norovirus on cruise ships, the best parameterization separates **clinical illness**, **infection**, and **detection**, because VSP-style attack rates capture reported clinical AGE rather than all infections. Cruise evidence shows substantial underascertainment from non-reporting, while outbreak, serologic, and molecular studies show a sizable asymptomatic reservoir and imperfect laboratory detection (Wikswow et al., 2011; Miura et al., 2018; Misumi & Nishiura, 2021; O'Reilly et al., 2026).

Clinical Ascertainment

A cruise-ship outbreak investigation found infirmiry surveillance captured only **60% of AGE cases**, because 40% of case passengers did not report to the infirmiry (Wikswow et al., 2011). Passenger behavior changed during the outbreak, with over 90% reporting increased hand hygiene and ill passengers reducing public activities more often than well passengers, so the public-health response likely alters the observation process after detection (Wikswow et al., 2011). More generally, outbreak surveillance systems are inconsistent in case definitions across sites, which matters if you are borrowing parameters across pathogens (Wikswow et al., 2021).

- **Observed clinical attack rate** on cruise ships should be modeled as reported symptomatic AGE, not total symptomatic AGE (Wikswow et al., 2011).
- A first correction is a **reporting multiplier** of about 1.67 for symptomatic cases on cruise ships (Wikswow et al., 2011).
- Behavior and control measures likely create **time-varying ascertainment** after outbreak recognition (Wikswow et al., 2011).

Asymptomatic Infection

Across outbreak-focused meta-analysis, PCR-positive asymptomatic norovirus prevalence among exposed people was **21.8%** globally and **17.6%** in China (Wang et al., 2023; Wang et al., 2024). A model-based reanalysis of 55 Japanese foodborne outbreaks estimated the **asymptomatic ratio among infections** at 32.1% overall and 40.7% for GII.4, while a larger Japanese time-series estimated 18.6% overall and as high as 25.8% for GII.4 in some years (Miura et al., 2018; Misumi & Nishiura, 2021). Serology-based modeling in English children inferred infection rates higher than disease incidence, consistent with substantial asymptomatic infection (O'Reilly et al., 2026).

- For norovirus, a defensible latent range for **P(asymptomatic | infected)** is roughly **0.19-0.32**, with higher values for **GII.4** (Miura et al., 2018; Misumi & Nishiura, 2021).
- PCR-positive asymptomatic prevalence among exposed cohorts is not identical to the asymptomatic ratio, because it mixes true silent infection with **post-symptomatic shedding** (Wang et al., 2023; Bucardo, 2018; Qi et al., 2018).
- Age and setting matter: asymptomatic prevalence peaks in **preschoolers** and the **elderly** in Chinese outbreak data (Wang et al., 2024).

Testing and Detection

qRT-PCR is much more sensitive than older methods and strongly shifts estimated silent infection burden (Bucardo, 2018; Khan et al., 1994). In a classic cruise-ship outbreak, PCR detected norovirus in **16 of 19 fecal specimens**, versus 7 by electron microscopy and 5 by antigen EIA, and paired serology showed a fourfold antibody rise in **10 of 12** tested people (Khan et al., 1994). Model-based work estimated the probability of virus detection from an infected person at **63.3%**, while another outbreak model treated symptomatic and asymptomatic infections as having similar shedding detectability under sensitive PCR (Misumi & Nishiura, 2021; Miura et al., 2018).

Measure	Model-ready estimate	Why it matters
Symptomatic reporting to infirmary	0.60	VSP undercounts symptomatic cases
Asymptomatic ratio among infected	0.19-0.32	Converts infection attack rate to clinical attack rate
PCR-positive asymptomatic prevalence in outbreaks	0.18-0.22	Useful for cross-sectional screening modules
Probability infected person tests positive	~0.63	Links latent infection to observed lab positives
GII.4 adjustment	higher asymptomatic fraction	Genotype-specific calibration

FIGURE 1 Norovirus parameters most useful for converting cruise-ship observed clinical attack rates into latent infection burden.

- Prefer **RT-qPCR** for stools, with Ct or viral load retained because symptomatic cases tend to have lower Ct values than asymptomatic carriers (Bucardo, 2018).
- Add **paired serology** when possible, because molecular assays miss some infections and serology captured immune responses during cruise outbreaks (Khan et al., 1994; O'Reilly et al., 2026).
- Wastewater surveillance on cruise ships appears useful for **early warning** and trend detection, with comparable NoV-GII performance between passive and composite sampling (Zafeiriadou et al., 2026).

A practical norovirus observation model for cruise ships is therefore: latent infections → symptomatic fraction → self-reporting fraction → lab-testing fraction, with genotype, age, and outbreak-response effects allowed to vary over time. For personal or operational health decisions, a licensed public health or clinical expert should interpret these parameters in the specific surveillance setting.

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